

eRIC-LR2021

LPRS eRIC-LR2021 LoRa/LoRaWAN-capable NTN module for OEM integration



Introduction

The eRIC-LR2021 radio module, powered by Semtech's fourth-generation LR2021 multi-band LoRa transceiver, is designed for seamless integration with the proven eRIC product family in a compact 20 mm x 15 mm form factor.

The eRIC-LR2021 is intended to provide OEMs with flexible wireless connectivity for both terrestrial and non-terrestrial applications, including devices using Low Earth Orbit (LEO) satellite services where available.

LPRS's implementation is designed to support hybrid terrestrial and non-terrestrial communications, using network-selection logic to choose the most appropriate communications path for the application.

Features

The eRIC-LR2021 module incorporates Semtech's LR2021 multi-band transceiver, enabling:

- Dual-mode operation across terrestrial LoRa/LoRaWAN and satellite NTN/SATCOM use cases, subject to firmware, regional spectrum access, and network availability.
- Support for regional sub-GHz operation including 433 MHz, 470 MHz, 490 MHz, 868 MHz, and 915 MHz bands.
- Additional high-band operation covering 1.6 GHz to 2.5 GHz for L/S-band NTN/SATCOM and 2.4 GHz ISM applications as indicated in the RF specification tables.
- Receive sensitivity down to -141.5 dBm under LoRa SF12/125 kHz conditions.
- Compact mechanical size of 15.24 mm x 20.32 mm x 2.8 mm.

Applications

eRIC-LR2021 is suitable for any device requiring long range low power operation. It is suited to battery-powered applications, sensors, actuators, and equipment intended for use in noisy or harsh environments, where a robust RF link is required, eRIC-LoRa provides a robust and reliable wireless engine for OEM's to take advantage of.

RAIL | SECURITY | FINANCE | ASSET TRACKING | INDUSTRIAL CONTROL | SENSORS | REMOTE MONITORING

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Document History

Issue	Date	Revision
1.1	December 2025	Provisional datasheet
1.2	January 2026	Initial release
1.3	March 2026	U.FL connectors added; PCB change
1.4	April 2026	Added break-out-board to Eval kits & Accessories

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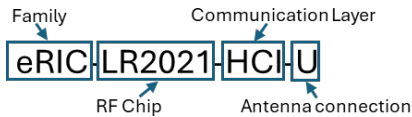
easyRadio modules are a component part of an end system product and should be treated as such. Testing to fitness and field testing is the sole responsibility of the manufacturer of the device into which easyRadio products are fitted. Any liability from defect or malfunction is limited to the replacement of product only and does not include labour or other incurred corrective expenses. Using or continuing to use these devices hereby binds the user to these terms.

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I. Module Formats.

The module is available in 4 formats as shown in the below table:



Order Code.	Description	HCI (Host Controller Interface SPI)	Standalone (UART – Data In / Data Out)	Antenna Connection
eRIC-LR2021-HCI-U	LoRa Dual Frequency LR2021 HCI Transceiver module for LoRaWAN & LoRa P2P applications. 2 x u.FL antenna connectors	✓	✗	u.FL/Pins
eRIC-LR2021-HCI-P	LoRa Dual Frequency LR2021 HCI Transceiver module for LoRaWAN & LoRa P2P applications 2 x pads/pins for antenna connection	✓	✗	Pins only
eRIC-LR2021-S-U	LoRa Dual Frequency LR2021 Standalone Transceiver module for LoRaWAN & LoRa P2P applications 2 x u.FL antenna connectors	✗	✓	u.FL/Pins
eRIC-LR2021-S-P	LoRa Dual Frequency LR2021 Standalone Transceiver module for LoRaWAN & LoRa P2P applications 2 x pads/pins for antenna connection	✗	✓	Pins only

I.1 eRIC-LR2021-HCI-x - Host Controller Interface Transceiver overview

The easyRadio eRIC-LR2021 transceiver module is a complete sub-system that combines a high-performance low power RF transceiver and a voltage regulator.

LPRS's eRIC series, featuring the LR2021, represents a fourth-generation “LoRa Plus” transceiver engineered to unify sub-GHz, 2.4 GHz ISM, and licensed S-band operation within a single, highly integrated radio platform. Designed for both terrestrial and NTN/SATCOM connectivity, it powers next-generation IoT and AI edge devices that demand ultra-low power consumption, extremely long range, and significantly higher data throughput than classic LoRa—while maintaining full backward compatibility with existing LoRa and LoRaWAN networks if fully supported by firmware, band plans, and protocol implementation for the intended variants.

I.2 eRIC-LR2021-S-x – Standalone UART Transceiver overview

The easyRadio eRIC-LR2021-S-x RF transceiver module is a complete sub-system that combines a high-performance low power RF transceiver, an ARM microcontroller and a voltage regulator.

The Serial Data Input (SDI) and Serial Data Output (SDO) operate at a standard 19,200 Baud rate, with two handshake lines available for optional flow control. The easyRadio Transceiver internally buffers data before transmitting it in an efficient over-air code format.

Any other eRIC-LR2021 transceiver, within range and on the same settings, that ‘hears’ the transmission will decode the message and place the recovered data within a receive buffer that can then be downloaded to the receiving host for processing and interpretation.

Radio transmission and reception is bi-directional (half duplex) i.e. transmit OR receive but not simultaneously. Extra internal buffers, however, allow the user to upload data while a download is in progress giving the appearance of fully duplex data flow. Commands for controlling these variants via UART communication—supporting both LoRaWAN and Point-to-Point connectivity—will be available from May 2026.

2. Pin/Pad Description

Pad No	HCI Version	UART Version	Description	Notes
1	LR2021 BUSY	GPIO/RSSI	General Purpose Digital I/O	Reserved
2	LR2021 SPI CLK	UART BUSY OUT	SPI Clock	Used in SPI mode
3	LR2021 SPI MISO	UART SDO	SPI data out (MISO); default receive data or Serial Data Out	SPI output - Connect to Host SPI input to receive wireless data.
4	LR2021 SPI MOSI	UART SDI	SPI data in (MOSI); default transmit data or Serial Data In	Connect to the host SPI output to transmit wireless data.
5	LR2021 SPI NSS	UART RTS	SPI select	Pull low to enable and select the SPI lines.
6	VCC	VCC	Operating Supply Voltage	Internal 3.3V regulator operates from +2.4V to +6V Input.
7	GND	GND	Power Ground	0V Ground
8	LR2021 VBAT	LR2021 VBAT	Regulator Output Measurement	
9	NRESET	NRESET	LR2021/MCU reset (LOW)	
10	N/C	SW CLK	STM32 Clock Programming	ARM MCU programming Pins
11	N/C	SW DIO	STM32 Data programming	
12, 13	N/C	N/C	Not connected	Not connected
14	LR2021 DIO6	LR2021 GPIO	LR2021 GPIO	User controlled direct from LR2021
15	LR2021 DIO5			User controlled direct from LR2021
16	LR2021 DIO9			User controlled direct from LR2021
17,18,19	N/C	GPIO	STM32 GPIO (PA8, PA9, PA10)	For "S" version (MCU) only
20	LR2021 DIO7	LR2021 GPIO	LR2021 GPIO	User controlled direct from LR2021
21	LR2021 DIO10			User controlled direct from LR2021
22	RF LF	RF Ground – 0V	433/868/915MHz sub-GHz antenna port. (Low frequency)	Connect to a suitable antenna via a 50-ohm PCB trace.
23	RF GND			Connect to antenna ground and the local ground plane; internally connected to power ground.
24	RF HF		1.5-2.5GHz antenna port. (High frequency)	Connect to a suitable antenna via a 50-ohm PCB trace.

Notes

3.3V Regulator will cap the maximum voltage to the RF IC. Internal RF IC can operate down to 1.8V and still provide rated RF power output.

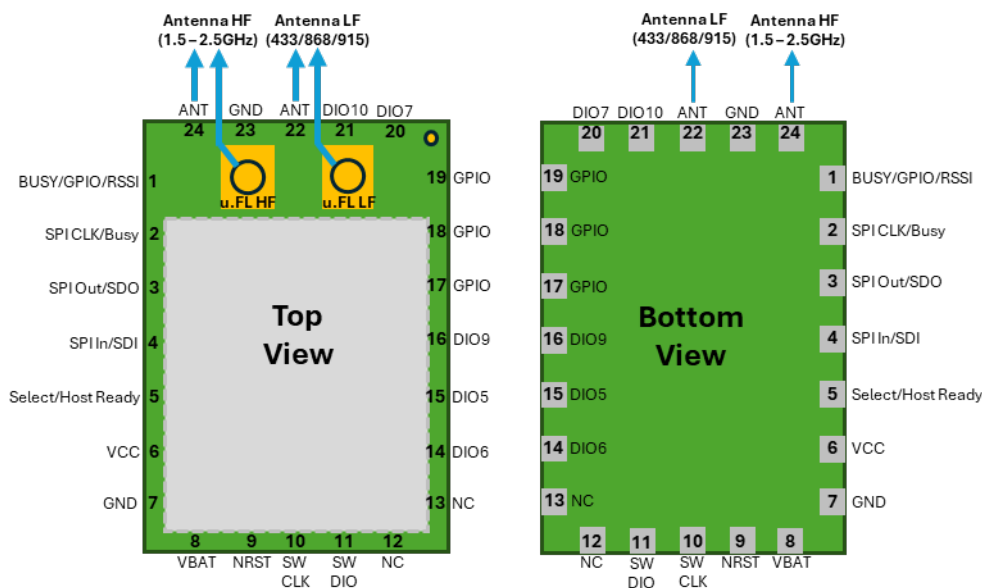


Figure 1 Pin pad identification.

3. Mechanical & Production

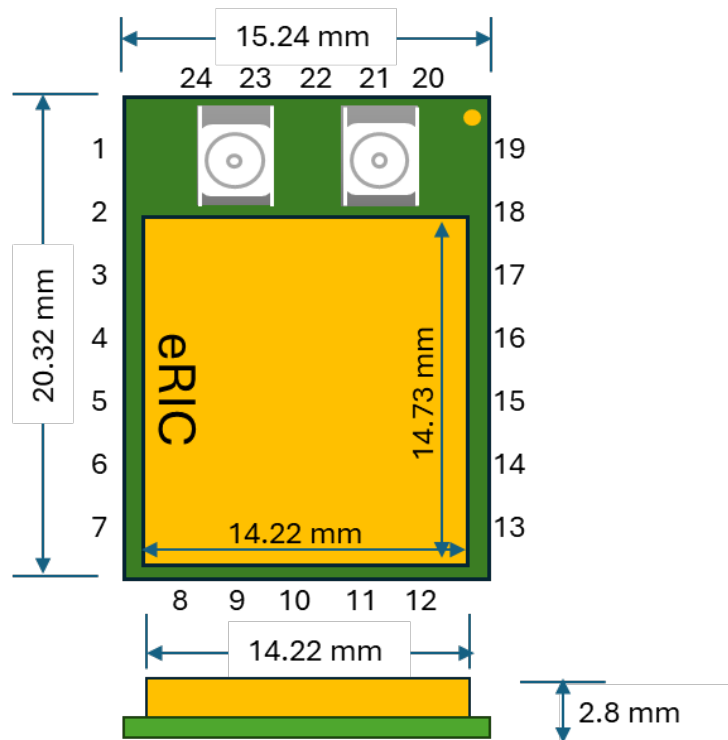


Figure 2 Mechanical Drawing

3.1 PCB Layout Notes

Pitch of the connection pads is 2.54mm. Pads 4 & 16 and 10 & 22 are on centre line (C/L) of module

It is recommended that the module is mounted on a double-sided PCB and that the area beneath the module be flooded with additional copper ground plane if possible. This ground area should be connected to pad 23 (RF Ground) and pad 7 (Power Ground).

The recommended pad layout is shown below. Pads should be solid and should not contain holes.

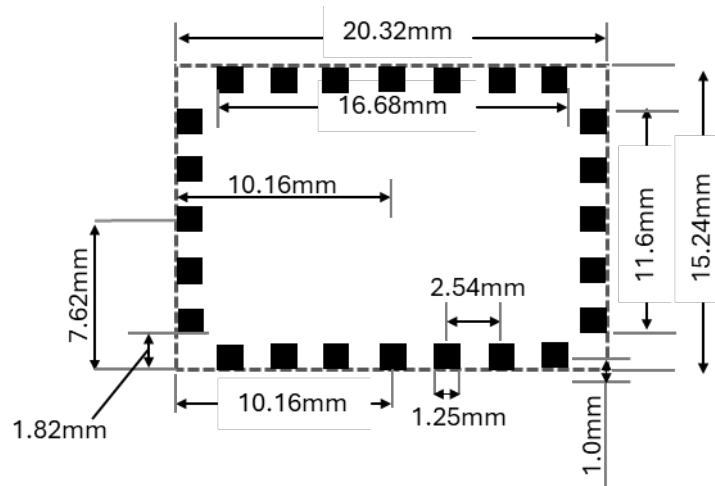


Figure 3 PCB Layout Dimensions

3.2 Production - Soldering & Reflow

eRIC modules are designed for reflow soldering but should only be subjected to the process once. Multiple passes may cause permanent internal component damage and are therefore not recommended.

Should multiple reflows take place, the customer does so at their own risk and LPRS provides no warranty against damages caused to this module.

For optimum reflow of this module, please see below our recommended reflow soldering profile for this device.

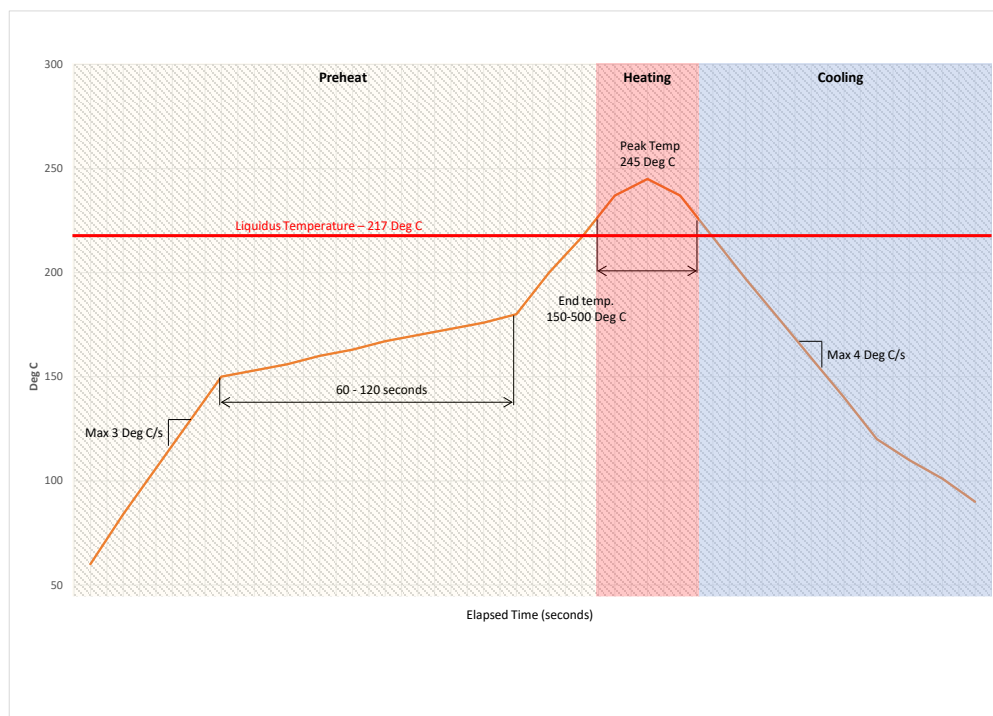


Figure 4 Soldering profile

Please note that whilst every care and attention has been made by LPRS to ensure this profile is correct, each manufacturing process is subtly different, and we recommend consulting with your manufacturing facility prior to production to ensure this profile is suitable for use with your overall PCB assembly process.

3.3 Soldering Paste & Stencil Thickness

LPRS recommend a solder past thickness of between 100 to 150 microns. Of course, each customer's specific implementation of the eRIC component is different and thus, the selected soldering paste thickness depends on the specific manufacturing requirements of the host printed circuit board and components used.

Please note that our statements relating to manufacturing procedures are recommendations only and not specifications. The exact solder, copper, paste mask dimensions, stencil thicknesses and solder paste quantity must be adapted to the specific production process of the customers designed PCB.

3.4 Conformal Coating.

Conformally coating LPRS's RF products may adversely affect the operation and RF performance of the device and therefore we would not recommend this course of action. If your application requires a coating of some kind, customers should conduct their own tests to ensure suitability. Whilst the RF shielding CAN will provide some protection, all care and attention should be taken to ensure the coating does not flow into the module.

Please note that conformal coating of the module with any substance will invalidate the warranty.

Applications that require conformal coating should be discussed with LPRS before release to production.

4. Absolute Maximum Ratings

Operating Temperature Range	-40° C to +85° C
Storage Temperature Range	-40° C to +85° C
Vcc	- 0.3 to + 6.0 Volts
All Other Pins (N.B.)	- 0.3 to +3.3 Volts

Performance Data:

The source table is stated for eRIC-LR2021 at +5.0 V supply $\pm 5\%$ and 20 C ambient temperature

DC Parameters	Pin	Min	Typical	Max	Units	Notes
Supply Voltage (Vcc)	6	1.8	3.3-5.0	6.0	Volts	Operating range per electrical table
Transmit supply current	6		120		mA	@+22dBm
	6		30		mA	@+14dBm
Receive supply current	6		5.7		mA	@sub-GHz
Low sleep current	6		470		nA	
Interface Levels						
Data Output Logic 1		1.8	3.1	3.1	Volts	
Data Output Logic 0			0.1		Volts	
Logic Output Current				25	mA	
Data Input Logic 1		2.0		3.6	Volts	
Data Input Logic 0				0.2	Volts	
RF Parameters	Pin	Min		Max		
Antenna Impedance	22, 24		50		Ohms	
Frequency Range	22	150	-	960	MHz	ISM 433, 470, 490, 868, & 915
	24	1500	-	2500	MHz	L/S band, NTN/SATCOM, ISM
Frequency Region	22	433.05	-	434.79	MHz	European ISM bands
		863	-	870	MHz	
		903	-	925	MHz	USA ISM bands
	24	1.5	-	2.5	GHz	Worldwide Frequencies
Transmitter						
RF Power Output				+22	dBm	Sub GHz
				+12	dBm	1.5-2.5GHz
Over Air Bit Rate				125	kbps	LoRa
				2.6	Mbps	FLRC Fast long-range communication
Receiver						
Sensitivity				-141.5	dBm	LoRa SF12/125KHz
Logic Timing						
Initial Power Up Time			TBA		mS	
Mechanical						
Size			15.24 x 20.32 x 2.8		mm	
Pin Pitch			2.54		mm	(Standard 0.1 Inches)
Weight			2		grams	

Notes *:

5. Logic Levels

The module operates internally from the output of an on-board 3.3 Volt low drop regulator. This regulator will still provide an (un-regulated) output below its drop out voltage, down to the minimum operating voltage (1.8V) of the RF IC.

The logic levels of the input/output pins are therefore between 0 Volt and the actual output voltage of the internal regulator. Outputs will drive external logic operating at 3.3 Volts. Resistors (1k typical) should be fitted in series with input data lines when interfacing to external logic outputs greater than 3.3Volts or greater than the external supply to prevent driving excess current into inputs and thus damaging them.

6. Power Supply

The supply used to power the transceiver should be 'clean' and free from ripple and noise (<20mV p-p total).

It is suggested that 100nF ceramic capacitors be used to de-couple the supply close to the power pins of the transceiver. The use of 'switch mode' power supplies should generally be avoided as they can generate both conducted and radiated high frequency noise that can be very difficult to eliminate.

This noise may considerably reduce the performance of any radio device that is connected or adjacent to such a supply.

For battery powered applications, we would recommend LPRS's range of 3.6V Lithium primary cells, available in all sizes including ½ AA, A, AAA, C, D, DD which range in capacity from 1.2aH to 19aH.

7. Antennas

The eRIC-LoRa transceiver can be used with the various common types of antenna that match the 50Ω RF Input / Output such as a monopole (whip), a tuned helical antenna, a PCB loop antenna or a ceramic 'chip' antenna.

Monopole antennas are resonant with a length corresponding to one quarter of the electrical wavelength ($\lambda/4$). They are very easy to implement and can simply be a 'piece of wire' or PCB track which at 868MHz should be approximately 8.6cm in length. This should be kept straight, in 'free space' and well away from all other circuitry, conducting objects and metalwork and should preferably be connected directly to the Antenna pin (24) of the transceiver.

If the antenna needs to be remote it should be connected via a 50Ω coaxial feeder cable or transmission line. A 50Ω transmission line can be constructed on FR4 board material by using a 3mm wide PCB track over a ground plane and this should be kept as short as possible.

The eRIC transceiver is also fitted with u.FL RF connector wired in parallel with pin 23 (RF Gnd) and pin 22 / 24 (RF In/Out). LPRS can supply suitable antennas fitted with matching connectors and low loss cable assemblies.

Helical antennas are also resonant and generally chosen for their more compact dimensions. They are more difficult to optimise than monopole antennas and are critical with regard to any surrounding conducting objects that can easily 'de-tune' them. They operate most efficiently when there is a substantial ground plane for them to radiate against.

PCB loop antennas are the most compact antennas but are less effective than the other types. They are also more difficult to design and must be carefully 'tuned' for best performance.

Chip antennas are attractive as they are compact and if used in accordance with the manufacturer's specifications can provide very good performance.

The Internet can provide much useful information on the design of Short-Range Device (SRD) Antennas.

Please Note: To meet US FCC requirements the modules must be used with the specified antennas (TBA) that were used for testing. Please refer to FCC documentation as available.

7.1 Antennas Approved for use with eRIC-LR2021

Part Number	Gain	Supplier	Notes
TBA			

8. Transmission restrictions

As with all devices using the ISM license free radio spectrum, LPRS's eRIC-LR2021 must comply with regional restrictions on spectrum access and use. Restrictions are usually limited to 4 main topics:

- 1) Operational frequency band (What frequency you are using to transmit data on).
- 2) Maximum effective radiated power (How much output power you are using to transmit).
- 3) Channel access (aka Duty Cycle, how long you are transmitting for each time you transmit and how often you are transmitting).
- 4) Maximum occupied bandwidth (How much of the band you can transmit over).

8.1 eRIC-LR2021 transmission guide

TBA

8.2 Programming Modules in Situ

LPRS recommends customers make provision for programming the modules firmware physically through their host MCU or separate programming port.

9. Product Order Code

9.1 Modules

Order Code	Description
eRIC-LR2021-HCI-U	LoRa Dual Frequency LR2021 HCI Transceiver module for LoRaWAN & LoRa P2P applications 2 x u.FL antenna connectors
eRIC-LR2021-HCI-P	LoRa Dual Frequency LR2021 HCI Transceiver module for LoRaWAN & LoRa P2P applications 2 x pads/pins for antenna connection
eRIC-LR2021-S-U	LoRa Dual Frequency LR2021 Standalone Transceiver module for LoRaWAN & LoRa P2P applications 2 x u.FL antenna connectors
eRIC-LR2021-S-P	LoRa Dual Frequency LR2021 Standalone Transceiver module for LoRaWAN & LoRa P2P applications 2 x pads/pins for antenna connection

9.2 Eval kits & Accessories

Order Code	Description
eRIC-LR2021-BOB	2.54mm pitch, break-out-board for eRIC-LR2021 module

Please contact the sales office for availability of other variants of the standard product. The software interface can be customised to specific requirements for high volume applications.

9.3 Firmware

eRIC-LR2021-x-x will be supplied with the latest release of firmware or a user specified version.

9.4 MTBF

The mean time before failure is >100 Years.

10. Packing – Trays, Tape & Reel.

10.1 Trays

eRIC's can be supplied in tray format as per the below image. Each tray holds a maximum of 40 x eRIC's and is 255mm x 155mm x 10mm.

Trays are designed to be stacked together in any quantity.

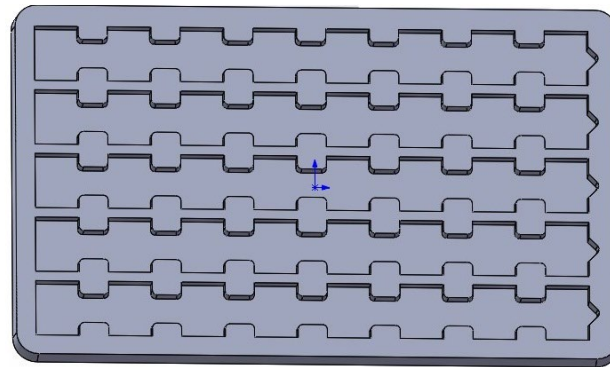


Figure 8. Drawing of each tray

10.2 Tape & Reel

eRIC's can be supplied taped and reeled in quantities of 150 modules per reel. The tape and reel is an industry standard with the following dimensions / specification:

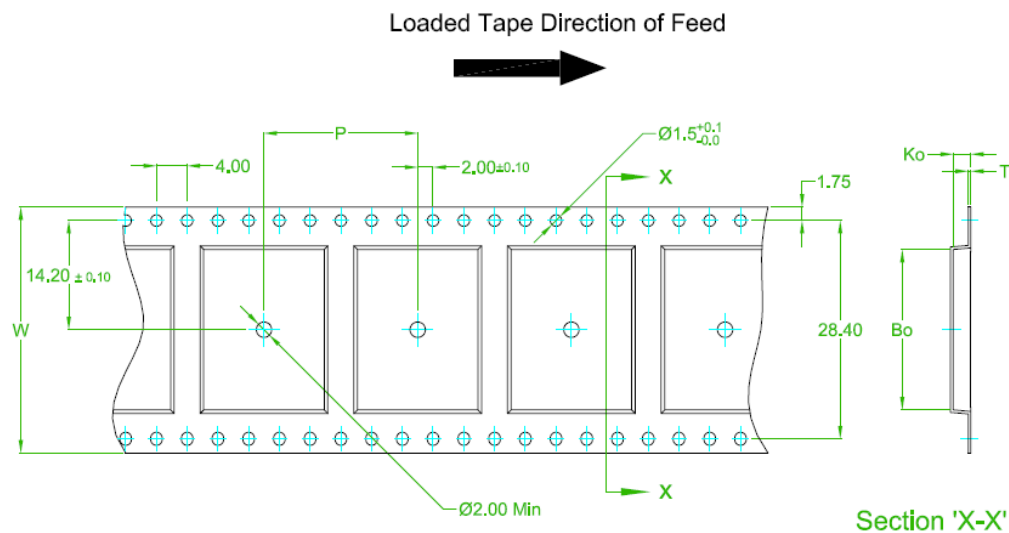


Figure 9. Tape and Reel dimensions

Tape & Reel Dimensions			
Ao	15.70 ±0.10	P	20.00 ±0.10
Bo	20.90 ±0.10	T	0.30 ±0.05
Ko	2.30 ±0.10	W	32.00 ±0.30
Please note: All dimensions in millimetres. 10 sprocket hole pitch cumulative tolerance ±0.20. Material: Conductive polystyrene. Camber not to exceed 1.0mm N 100mm			

11. Regulatory Approvals

In progress

11.1 FCC Statement:

In progress

12. eRIC-LR2021 Firmware Versions

Version	Date	Revision	Known Issues